

Note: Some students may not realise that the reaction produced an increase in mass.

- 7 (a) (i) Work done W by a force F is the product of the force and the distance r moved in the direction of the force, $W = Fr$

The total work done by a variable force will thus be $W = \int F dr$.

The potential energy E_p is equal to the work done by an external agent in bringing the two ions together from infinity. The force exerted by the external agent is opposite to that of the electric force.

Hence, $E_p = - \int F dr$

$$F = - \frac{dE_p}{dr}$$

$$= -G$$

$$|F| = G$$

- (ii) For $r < 2.8 \times 10^{-10}$ m, the gradient $\frac{dE_p}{dr}$ is negative.

The force $F = - \frac{dE_p}{dr}$ is positive, indicating a repulsive force.

For $r > 2.8 \times 10^{-10} \text{ m}$, the gradient $\frac{dE_p}{dr}$ is positive.

The force $F = -\frac{dE_p}{dr}$ is negative, indicating an attractive force.

Note: A common error by students was to relate the sign of the potential energy to the force rather than the sign of the gradient.

(iii) 1. At $r = 2.8 \times 10^{-10} \text{ m}$, the gradient of the graph is zero. Hence the force $F = 0 \text{ N}$.

2. At $r = 5.0 \times 10^{-10} \text{ m}$,

the gradient of the graph is $\frac{(6.0)(1.60 \times 10^{-19})}{(8.0 - 3.0) \times 10^{-10}}$

$$F = 1.9 \times 10^{-9} \text{ N}$$

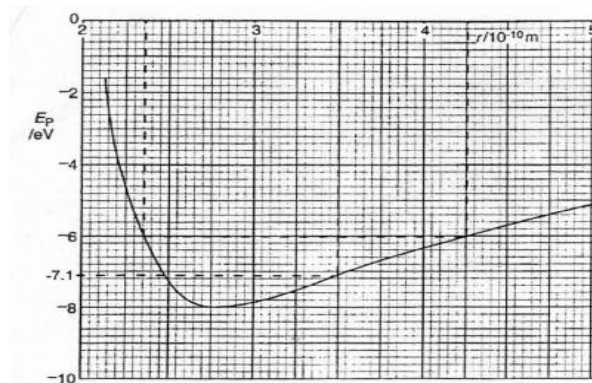
(b) (i) 1. The force causes repulsion between the ions.

This is because the gradient of the graph of the equation $E = \frac{B}{r^8}$ is negative,

hence $F = -\frac{dE_p}{dr}$ is positive.

2. The force is significant when the ions are very close to each other, but decreases rapidly with distance.

(c) (i)



1. Minimum value of $r = 2.35 \times 10^{-10} \text{ m}$. Maximum value of $r = 4.25 \times 10^{-10} \text{ m}$.

2. At distance $r = 3.5 \times 10^{-10} \text{ m}$, the potential energy of the ions is -7.1 eV .

$$\begin{aligned} \text{Kinetic Energy} &= \text{Total Energy} - \text{Potential Energy} = -6.0 \text{ eV} - (-7.1 \text{ eV}) \\ &= 1.1 \text{ eV}. \end{aligned}$$

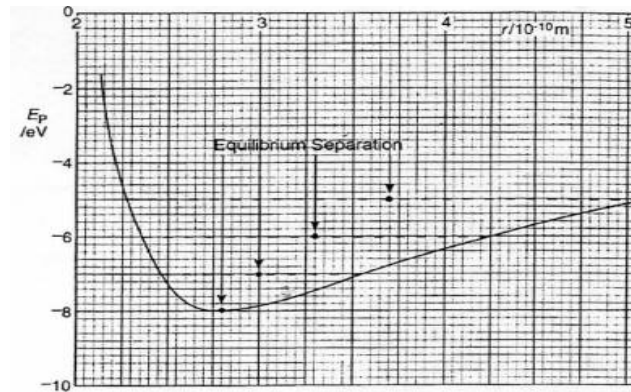
(ii) The motion is not simple harmonic because the force F is not directly proportional to displacement r (in fact the graph is not even symmetrical about the equilibrium

position at $r = 2.8 \times 10^{-10} \text{ m}$). Since acceleration $a = \frac{F}{m}$, the acceleration of the ions is not directly proportional to displacement r .

This violates one of the two conditions for simple harmonic motion (which is that the acceleration of the body must be directly proportional to its displacement from the equilibrium position).

- (d)** As the lattice is heated, its total energy increases. The total energy of the ions intersects the graph at higher levels. As a result, the equilibrium position (mid-point) of the vibration becomes progressively greater. The equilibrium positions of four different energy levels are shown:

With a greater equilibrium separation, the lattice expands.



Total Energy / eV	Equilibrium Separation $r / 10^{-10} \text{ m}$
-8.0	2.8
-7.0	3.0
-6.0	3.3
-5.0	3.7

Note: Common errors include referring to an increase in kinetic energy causing expansion with no reference to the changes in potential energy.